

Automated Forensic Detection of Injection Tampering in CT Imaging via MedSAM2-Enabled Semantic Segmentation

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Abstract

Injection-based tampering of medical images poses a great threat to the integrity of medical diagnosis and decision-making. Recent works in generative models, such as GANs and diffusion can easily insert or remove pathologies in volumetric CT scans which cannot be detected easily, even by expert radiologists which leads to false diagnosis. Existing deepfake detectors largely focus on just providing if the slice is authentic or not limiting their use in forensic and clinical fields. This work bridges that gap by using a two-stage framework for CT tamper detection and localisation, it combines a lightweight classifier along with a segmentation model. In the first stage ResNet-18 is used to determine whether the CT slices are tampered with or not, only the flagged slices are passed onto the MedSAM2 (a promotable segmentation model originally developed for medical 3D image segmentation) which localises the tamper location. This framework reduces computational costs while providing accurate localization. This proposed system is evaluated using publicly available CT subset of deepfake dataset containing GAN and diffusion-based injections. On this dataset the ResNet classifier achieves 0.991 accuracy on the training subset and 0.990 accuracy on the held-out test subset, with precision recall and F1-score all around 0.99. Qualitative experiments also show that MedSAM2 produces correct segmentation marks that align with the injected locations in both single cell and multi cell volumetric slices. These results show that this relatively simple and efficient system can produce effective tampering analysis which can be easily interpreted for forensic and clinical purposes.

Keywords: Artificial intelligence in radiology, Deepfake detection, Medical image forensics, Promptable segmentation, Two stage hybrid classification MedSAM2, ResNet, volumetric segmentation

1 INTRODUCTION

Medical imaging is the non-invasive process of producing internal visuals of a body for medical analysis, examination, and treatment. Volumetric (3D) scans are required to diagnose certain conditions. One of the most common techniques for producing detailed 3D medical imagery is Computed Tomography (CT). In 2023, approximately 93 million CT scans were performed in the US alone.

CT scanners are managed through a Picture Archiv-

ing and Communication System (PACS), which stores images digitally in a centralized repository, retrieves them as needed, and shares them with authorized users via a secure network. The digital image data follows the universal DICOM standards for storage and transfer.

The security of healthcare systems has been lagging behind [1-4]. This is partially because healthcare security policies mostly target data privacy instead of data security [1]. Some PACS are intentionally or accidentally exposed to the internet via web access solutions. Researchers at

tor. The generator generates fake samples with the aim of tricking the discriminator while the discriminator learns to classify whether the sample was synthesized or authentic. As a result, the system is capable of generating fake samples which are almost undetectable. This is demonstrated by Mrisky et al. [7] with their introduction to the CT-GAN framework. The framework uses two conditional GANs (cGAN) to perform in-painting on 3D imagery, for injection and removal of medical conditions. One cGAN (used for injection) is trained on unhealthy samples of volumetric scans so that the generator will always complete the image accordingly whereas, for removal, another cGAN is trained on healthy samples. In this paper, we evaluate the performance of MedSAM2 [8], a ‘Segment Anything Model 2’ (SAM2) [9] fine tuned for medical images. MedSAM2 is capable of handling both volumetric medical scans and successive video frames, enabling segmentation across diverse medical data. The MedSAM2 inference is run on the BTD-mri and ct deepfake test sets dataset [10].



Recent years have witnessed remarkable progress in both general-purpose segmentation foundation models and medical image forensics, giving rise to a growing intersection between segmentation, anomaly detection, and authenticity verification in medical imaging. This section reviews three major research directions that form the foundation for our proposed pipeline: (1) foundation-model-based medical image segmentation, (2) medical deepfake and image tampering detection, and (3) general image and video forgery detection approaches. Together, these domains contextualize the motivation behind combining classification-based screening with MedSAM2-guided tampering localization.

2.1 Foundation Models for Medical Image Segmentation

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a prompt-based segmentation workflow significantly reduced the annotation burden in clinical workflows. Building on this foundation, Wang et al. [8] introduced MedSAM2, a memory-augmented transformer architecture capable of handling 3D volumetric data and slice sequences by leveraging cross-slice temporal attention. This approach achieved consistent performance across heterogeneous medical imaging modalities, marking an important step toward unified segmentation frameworks.

Subsequent works have sought to improve efficiency and representation flexibility. I-MedSAM [13] integrated implicit neural representations (INR) into the SAM backbone, providing continuous and differentiable spatial representations that enhance fine boundary delineation while maintaining cross-modality generalization. Similarly, Rep-MedSAM [14] introduced lightweight representation bottlenecks to reduce computational complexity during inference, targeting deployment in clinical settings with limited hardware resources. Other extensions, such as SAM-AutoMed and SAM-MedCls [15], replaced or augmented the original prompt encoder with mobile-scale backbones, enabling fully automated segmentation pipelines that no longer rely on manual point or box prompts. These developments collectively highlight a paradigm shift in medical imaging, from dataset-specific CNN models to promptable foundation architectures that generalize across modalities and anatomical regions.

While these models excel in delineating anatomical structures or pathological regions, they typically assume that the underlying data is authentic. Few works have explored using foundation segmentation models for forensic purposes, such as isolating manipulated or fabricated regions within medical images. This gap motivates the present study, which integrates MedSAM2 not as a diagnostic segmentation tool but as a tampering localization engine operating downstream of a classification-based authenticity gate.

2.2 Medical Image Forgery and Deepfake Detection

Generative models ranging from conditional GANs to diffusion models has raised concerns about the integrity of medical imagery used for diagnosis, publication, and training. Early attempts, such as the study by Solaiyappan et al. [16], evaluated texture- and noise-based descrip-

tors, including local binary patterns and co-occurrence matrices, for discriminating between genuine and GAN-synthesized medical scans. Although effective under controlled conditions, these handcrafted methods lack robustness against novel generative architectures or unseen manipulation types.

More recent deep learning-based approaches have attempted to address these limitations. Alsabbagh and Al-Kadi [17] performed a comparative analysis of thirteen convolutional architectures, including ResNet, DenseNet, EfficientNet, and MobileNet, to assess their suitability for detecting medical deepfakes. Their results demonstrated that while deeper networks achieved higher accuracy, lightweight models offered more consistent generalization across modalities. Li et al. [18] introduced MedForensics, a large-scale benchmark dataset for medical deepfake detection and proposed the Dual-Stage Knowledge Integration (DSKI) framework, which combines spatial texture cues with frequency-domain residuals to identify manipulations in X-ray and CT imagery. Similarly, Grabovski et al. [19] proposed Back-in-Time Diffusion, an unsupervised approach that reconstructs input images through reverse diffusion and quantifies reconstruction residuals as indicators of tampering. Their method demonstrated strong detection performance on both GAN- and diffusion-generated medical deepfakes.

Denoising Diffusion Probabilistic Models (DDPMs) generally are used to generate images but they can also be used to detect anomalies. The general approach to this is, first the DDPM is trained on authentic data, then some noise is introduced to the target image, the DDPM is then tasked to denoise the target image, finally, the denoised image is compared to the original image. If the difference between the residuals of both the images is high then the original image is considered to be anomalous or tampered. However, this approach can lead to misleading results with production of semantically irrelevant noise that could lead to false positives. Often the magnitude of the added content (noise) would overpower the subtle forensic patterns discovered in the residue. Grabovski et al introduced BTD (Back-in-Time Diffusion) to counter these drawbacks. BTD involved directly denoising the image without adding any noise. As a result BTD's residue only contains forensic content.

A complementary line of work explores hybrid methods that combine spatial and spectral information. The

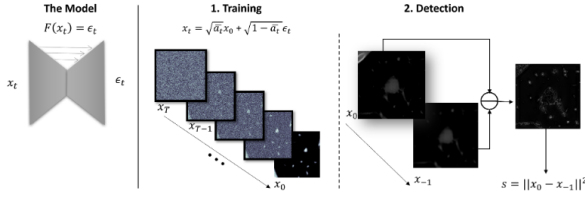


Figure 2: Overview of the BTD framework

Hybrid Spatial-Frequency Forensics Network [20] fuses CNN-based spatial encoders with Fourier-domain anomaly extractors to improve robustness against high-fidelity manipulations that preserve low-frequency structures. Despite these advances, most existing detectors are designed to classify images as “real” or “fake” without providing fine-grained localization of the manipulated regions. In medical contexts, however, localizing the tampered area is essential for ensuring interpretability and clinical trustworthiness. This distinction underpins the motivation for a two-stage approach, wherein a classifier first identifies potentially tampered slices, and a segmentation foundation model subsequently performs precise localization.

2.3 General Image and Video Forgery Detection

Beyond the medical domain, the broader field of digital forensics has produced numerous strategies for detecting image and video forgeries. Traditional methods rely on feature inconsistencies, such as compression artifacts, lighting mismatches, or camera model fingerprints. With the advent of deepfakes, neural network-based approaches have become predominant. Lee et al. [21] introduced the Transfer-Adaptive Residual (TAR) framework, a weakly supervised detector that learns manipulation-invariant features across multiple deepfake datasets. Other techniques leverage physiological signals, such as subtle facial pulse variations in DeepFakesON-Phys [22], to identify synthetic content. While these approaches target human faces rather than medical data, they highlight an important trend: the integration of domain priors with data-driven detection. A comprehensive survey [23] categorizes deepfake detection methods into spatial, frequency, and temporal domains, emphasizing that hybrid architectures, which

combine complementary cues, consistently outperform single-domain methods. However, such methods are seldom adapted for grayscale or volumetric data typical of CT or MRI scans, leaving a critical research gap for medical forensic applications.

3 PROPOSED METHODOLOGY

This work focuses on detecting and localizing injection-based tampering in volumetric CT scans. The focus is the locate the synthetic malignant tissues in otherwise healthy scans. The goal is to provide both:

- Slice-level authentication
- Pixel-wise localization of tampered regions

While existing medical deepfake detection methods can meet only the first requirement of detecting if a slice is tampered with, conventional segmentation methods assume that the input is authentic and do not consider forensic use cases focusing on verifying the integrity of the protected images and not on analysing the already compromised data. In contrast to the prior medical deepfake detection methods that focus on the image-level classification, and other segmentation methods that assume authentic inputs, this work addresses both using a two-stage approach.

The proposed framework:

- ResNet-18 model to authenticate the slice
 - MedSAM2 as a general-purpose segmentation to localize tampered regions
 - Targets forensic and clinical interpretation by integrating these two components in an adaptive routing pipeline
- The design intentionally relies on widely used architectures to maintain reproducibility.

The novelty of this research lies in its hybrid framework that integrates image level classification and pixel level localization. The proposed system introduces an adaptive routing mechanism, where only the classifier-flagged suspicious scans are passed to the foundation segmentation model (MedSAM2) for detailed analysis. This selective inference design not only reduces computational cost but also provides interpretability by providing visual evidence of manipulation. Furthermore, the work represents one of the first attempts to repurpose segmentation foundation models for forensic analysis, transforming MedSAM2 from a diagnostic segmentation tool into a tampering localization engine.

Table 1: Tabular comparison of advantages and limitations of related works

Work	Objective/Application	Advantages	Limitations
MedSAM [12]	Adapt SAM for medical segmentation	Cross-modality generalization; minimal fine-tuning	2D only; assumes authentic inputs
MedSAM2 [8]	Extend SAM to volumetric 3D segmentation	Handles slice sequences; high robustness	High compute cost; not designed for forensics
I-MedSAM [13]	Use implicit neural representation for SAM	Enhances fine detail; spatial continuity	Complex training; high memory demand
Rep-MedSAM [14]	Lightweight SAM variant for real-time use	Fast and efficient	Slight accuracy drop; small datasets
SAM-AutoMed / MedCls [15]	Automated segmentation and classification	Removes need for prompts	Limited interpretability; weak forensic insight
Solaiyappen et al. [16]	Texture-based fake detection	Simple; interpretable	Poor generalization; no localization
Alsabbagh & Al-Kadi [17]	Comparative CNN analysis	Comprehensive benchmarking	Binary detection only
Li et al. [18]	MedForensics dataset, DSKEI model	Spatial + frequency fusion	Limited modalities
Grabovski et al. [19]	Back-in-Time diffusion detection	Unsupervised; robust residuals	Expensive; noise-sensitive
Hybrid Spatial-Frequency [20]	CNN + Fourier cues	Robust to hybrid fakes	High training complexity
Lee et al. [21]	TAR generalized deepfake detector	Cross-domain transferability	Large training data required
Ciftci et al. [22]	Physiological deepfake detection	Uses biological cues	Not applicable to CT images

3.1 Classification

The first stage of the proposed pipeline focuses on determining whether an image is tampered or not, with the help of a binary classifier. Let each CT slice be denoted as a grayscale matrix

$$I \in \mathbb{R}^{H \times W} \quad (1)$$

where H is the height of the slice and W is the width of the slice. Since convolutional neural networks (CNNs) [24] such as ResNet [25] are pretrained on natural RGB images, the slice undergoes the following transformations:

1. Intensity Normalization: To standardize the heterogeneous scanner outputs, we apply either window-leveling, to map Hounsfield Units (HU) to normalized image pixels, or z-score normalization:

$$I' = \frac{\text{clip}(I, W_{\min}, W_{\max}) - W_{\min}}{W_{\max} - W_{\min}} \quad (2)$$

2. Spatial Resampling: Each slice is resized to a fixed spatial resolution $H' \times W' = 224 \times 224$ to match the CNN input size.

3. Channel Replication: Since I' has a single channel, it is replicated across three channels:

$$X = \text{stack}(I', I', I') \in \mathbb{R}^{3 \times 224 \times 224} \quad (3)$$

4. Data Augmentation: During training, transformations such as random flips, rotations and noise injections are applied to improve generalization and robustness against scanner/domain variability.

ResNet-18 is used as the convolutional backbone, initialized with ImageNet weights. The final fully connected layer is replaced with a binary classification head:

$$f_0 : \mathbb{R}^{3 \times 224 \times 224} \rightarrow [0, 1] \quad (4)$$

where $f_0(X)$ outputs the probability of tampering. The prediction for a slice is then:

$$\hat{y} = \begin{cases} 1 & \text{if } f_0(X) \geq \tau \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

with threshold τ typically set to 0.5. The Binary Cross-Entropy (BCE) loss is used:

$$L_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (6)$$

where $y_i \in \{0, 1\}$ is the ground-truth label (0 = *authentic*, 1 = *tampered*).

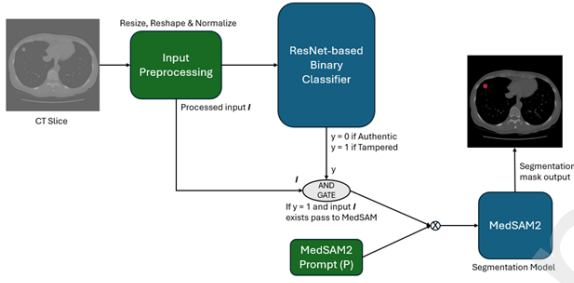


Figure 3: Proposed architecture diagram

3.2 Segmentation

Upon classification of an image slice as tampered in the classification stage, the pipeline advances to pixel-level localization to highlight the manipulated regions. This step leverages a promptable segmentation foundation model for 3D image and video segmentation MedSAM2. This model was developed by fine-tuning the ‘Segment Anything Model’ (SAM). The aim is to not only classify but also spatially delineate regions of tampering, providing interpretability and actionable evidence for forensic or clinical workflows. The input is the suspected tampered slice $I \in R^{H \times W}$. The preprocessing steps are consistent with those in the classification stage to ensure compatibility:

1. Normalization: The intensity is scaled using window-leveling (for mapping HU to normalized image pixels) or z-score normalization.

2. Resizing: The slice is resized to match the expected resolution of the segmentation model $H'' \times W'' = 224 \times 224$.

3. Conversion to 3D: Neighbouring 2D slices are stacked to provide volumetric context to form a 3D image, $V = \{I_k - 1, I_k, I_k + 1\} \in R^{3 \times H'' \times W''}$.

The segmentation backbone (MedSAM2) uses a transformer-based encoder-decoder to propagate bounding-box or point prompts through the image and generate a segmentation mask.

$$M = g_\phi(I, P) \quad (7)$$

where $M \in \{0, 1\}^{H'' \times W''}$ is the predicted binary mask, g_ϕ is the segmentation model with parameters ϕ , and P denotes optional prompts (e.g. bounding boxes centered on suspicious coordinates).

Algorithm 1 Classification of CT image and segmentation

```

1: procedure PREPROCESS( $I$ )  $\triangleright$  Perform preprocessing
2:    $I' \leftarrow \frac{\text{clip}(I, W_{\min}, W_{\max}) - W_{\min}}{W_{\max} - W_{\min}}$ 
3:    $I'' \leftarrow \text{resample}(I', 224 \times 224)$ 
4:    $I''' \leftarrow \text{stack}(I'', I'', I'') \in R^{3 \times 224 \times 224}$ 
5:    $I_{\text{preprocessed}} \leftarrow \text{augment}(I''')$ 
6:   return  $I_{\text{preprocessed}}$ 
7: end procedure
8: procedure CLASSIFY( $I_{\text{preprocessed}}$ )  $\triangleright$  Classify as
   tampered or authentic
9:    $f \leftarrow \text{ResNet-18}$ 
10:  if  $f(I_{\text{preprocessed}}) \geq 0.5$  then
11:    return 1
12:  else
13:    return 0
14:  end if
15: end procedure
16: procedure SEGMENT( $I_{\text{preprocessed}}, P$ )  $\triangleright$  Run only if
   CLASSIFY returns 1
17:    $M \leftarrow \text{MedSAM2}(I_{\text{preprocessed}}, P)$ 
18:   return  $M$   $\triangleright$  Final segmentation mask
19: end procedure

```

4 EXPERIMENTAL SETUP

Datasets

The BTD-MRI and CT Deepfake Test Sets, from which all experiments were generated, and available on Kaggle [[10]]. In this paper, we are only dealing with the CT subset. This dataset comprises fully synthetic examples of medical imaging with injected malignancies, both GAN-based and stable-diffusion based, that were generated. The modifications are spatially confined, which allows for both image-level detection and pixel-level isolation to be assessed.

Approximately 3,900 CT slices were collected, as detailed in the result evaluation section. The dataset contains paired genuine and manipulated instances, but the dataset meta-data does not explicitly specify the ratio distribution of GAN and diffusion injected samples, nor the class distribution of genuine and tampered samples. Therefore, we consider the dataset as a mixed forgery benchmark for forensic analysis.

Data Splits

The data was split into training and test sets, 80% as the training set and 20% as the test set. This was in relation to the procedures described in the results section. The training set was used to train the model, while the test set was used to test the trained model for its final performance. The same split was applied to provide fair comparisons in all experiments. There was no use of validation set and cross-validation.

Preprocessing Details

Highly intense areas in the CT scan slices should be normalized through Hounsfield Unit (HU) windowing or z-score normalization. The normalization process can reduce the magnitude of the intensity variance differences among the CT scan machines and can result in the improvement of the generalizations of the model. The HU values for the study are in the range of $[-160, 240]$, and those outside this range are removed.

Additionally, to make predictions, the images are bilinearly resampled to 224×224 . Since the ResNet-18 requires the image to consist of three channels, the grayscale images are replicated for three channels. Similarly, to perform the segmentation process, the images are downsampled to 512×512 . Finally, adjacent 2D images for each slice are stacked to form a 3D image to provide the MedSAM2 with local 3D context.

Data augmentation (training only)

To increase generalization and robustness, we applied data augmentation only on the training dataset, which comprised random rotation of images by $\pm 15^\circ$, flip operations along both horizontal and vertical axes, possibility of elastic distortions, and addition of noise with $\sigma = 0.01$ to the images. All these are aimed at ensuring that the images are viewed from different structural semantic perspectives.

Hyperparameters

“Classification backbone” (ResNet-18) was trained using pre-extracted weights from ImageNet. The Adam optimizer was used to train this model with a learning rate of 1×10^{-4} and exponential decay of 0.95 per epoch. Batch size was fixed at 32, and the maximum number of epochs was 100 with early stopping having a patience of 10. Binary cross-entropy (BCE) loss function was used with a decision threshold of 0.5. Class weight was not mentioned.

Segmentation Prompting

Segmentation was performed using a promptable base segmentation model (MedSAM2). The bounding-box prompts were automatically generated based on the regions of interest spuriously identified by classifier with Grad-CAM activation maps. We emphasize that each image was presented with the use of a single prompt, and shifted dimension was only slightly jittered to achieve additional robustness.

Evaluation Metrics

The classification performance is evaluated using the Accuracy, Precision, Recall, and F1 score. Quality of the segmentation: As a quantitative evaluation on the quality of the segmentation, we provide predicted mask visualizations.

Computational Environment

All our testing is conducted on a workstation equipped with an NVIDIA RTX 4060 Graphics Card (12GB of VRAM). The end-to-end inference time was $\approx 12s$ /CT volume. The time taken for the training procedure was not formally documented.

5 TEST RESULTS

The dataset used for this experiment is the CT subset of the dataset: BTD-MRI and CT deepfake test sets from kaggle. The subset consists of GAN induced injection and stable diffusion based injection of fake-malign can-

cer to CT scans. Approximately 3.9k samples were used to evaluate the performance of our proposed workflow. The entire experiment was run on an RTX 4060 GPU hence proving the computational efficiency of the workflow. The preprocessing steps mentioned in the previous section were used, namely, Intensity Normalization, Spatial Resampling, Channel Replication, and Data Augmentation. t2 shows the performance of the classifier during training and testing. f3 shows the confusion matrix thereby further explaining the performance of the classifier over the test subset. The segmentation results using MedSAM2 are shown in f4.

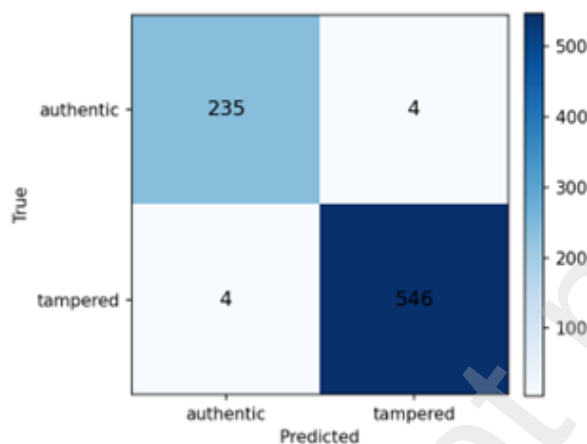


Figure 4: Confusion matrix depicting the performance of the classifier (calculated over the test subset)

6 CONCLUSION

This work presents a combined framework for medical image authenticity verification and tampering localization, combining a deep classification model with a foundation-level segmentation model. The ResNet-based classifier used in the second stage (classification stage) efficiently filters out authentic and tampered medical scans and isolates only potentially manipulated scans for further analysis. The third stage leverages MedSAM2, a promptable segmentation model originally designed for general medical image segmentation, to localize tampered regions with



Figure 5: MedSAM2 segmentation results, left image is the tampered image and middle image is the MedSAM2 segmentation mask output and the right image is the coordinates of the region of interest (as per dataset)

high precision. This selective routing approach not only minimizes unnecessary computational overhead but also enhances interpretability by offering both image-level and pixel-level forensic evidence. Experimental evaluations demonstrate that the proposed pipeline achieves high accuracy in detecting forged CT slices, with the segmentation module successfully localizing manipulated regions corresponding to injected pathologies. The modular design of the system ensures adaptability across various modalities and manipulation types, suggesting that the same framework can be extended to MRI or ultrasound data without significant re-engineering. Importantly, the classifier-segmentor coupling introduces a practical balance between efficiency and explainability, aligning with clinical requirements for transparent and interpretable decision-making. Future research will focus on integrating temporal and volumetric coherence to extend the method to full 3D series, exploring diffusion-based priors for improved anomaly sensitivity, and developing multi-modal extensions that incorporate radiology reports or metadata for holistic authenticity assessment.

Table 2: Classifier performance metrics, the dataset was split into Train subset (80% of data) and Test subset (20% data)

Subset	Accuracy	Precision(P)	Recall(R)	F1-Score
Train	0.991	0.991	0.995	0.993
Test	0.990	0.993	0.993	0.993

7 FUTURE WORKS

We will concentrate on the robustness, generalization capabilities, and clinician suitability of the proposed framework in our future endeavours. First, our intention is to test our framework on patient-derived tampering cases. The existing results are only on simulated tampering cases. The validity test will give us a chance to test our framework on patient-derived cases that have clinical-level variations. Second, our goal would be to make our framework more extensible to new imaging modalities. Notably, new imaging modalities such as MRI and ultrasound are very differing in image quality compared to the existing CT imaging technology. We will determine whether domain adaptation-based fine-tuning or new prompt engineering approaches for preserving the localization results are possible in our framework. Third, our goal will be to upgrade our threat model by incorporating diffusion-based forgeries, inpainting attacks, or copy-paste-based manipulations. We will attempt to combine our threat level by utilizing uncertainty-based inference and adversarial training. Finally, our goal will be to validate our framework in a PACS environment through latency difference compression. We will finally put our efforts into doing a clinician-level user study analysis for better interpretability of our framework results.

Author Contributions

Conceptualization, Aruna S and Vimaladevi; methodology, Aruna S; software, Aruna S and Vimaladevi; validation, Aruna S, Vimaladevi, Surabhi Narayan; formal analysis, Aruna S and Surabhi Narayan; investigation, Aruna S; resources, Aruna S; data curation, Aruna S; writing—original draft preparation, Aruna S and Vimaladevi; writing—review and editing, Aruna S and Vimaladevi; visualization, Surabhi Narayan; supervision, Surabhi Narayan; project administration, Surabhi Narayan.

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Declarations

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Ethics Approval

I/We declare that the work submitted for publication is original, previously unpublished in English or any other language(s), and not under consideration for publication elsewhere.

Consent to participate / Informed Consent

Not Applicable.

Consent for publication

I certify that all the authors have approved the paper for release and are in agreement with its content.

Data Availability

The datasets analyzed during the current study are available in the repository, <https://www.kaggle.com/datasets/freddiegraboski/btd-mri-and-ct-deepfake-test-sets>

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